Greedy 3-Sum-Free Sequences in Regular, Boundary, and Dense-Cap Families

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Abstract

We study the greedy sequence $S_{1,y,z}$ obtained by beginning with the three seeds $1, y, z$ and adjoining each later integer that is not a sum of three distinct earlier terms. For each $h = y + 1$ and rank $r \geq 1$, we give exact eventually periodic descriptions in three regions of the third-seed parameter: a regular band, its upper boundary point, and the dense portion of the following transition window. In each recurrent block of $2h$ third-seed values these results classify

$$2h - 2 - \left\lceil \frac{h+2}{2} \right\rceil$$

values, so the classified proportion tends to $3/4$ as $h \to \infty$. The rank-one regular case resolves Conjecture 6 of Bosma, Bruin, Fokkink, Grube, Reuijl, and Tromp, and extends its formula to the adjacent endpoint. The proofs use a seeded safety-and-coverage criterion, restricted sums of integer intervals, and modular tail-safety arguments.

1 Introduction

Let $x < y < z$ be positive integers. The greedy 3-sum-free sequence $S_{x,y,z}$ is the increasing sequence which begins with x, y, z and then, for each later integer, adjoins the integer exactly when it is not a sum of three distinct earlier terms of the sequence. We identify the increasing sequence with its set of terms, so membership statements such as $n \in S_{x,y,z}$ are used throughout.

A local greedy rule of this kind can nevertheless settle into a rigid eventually periodic pattern. For example, Theorem 4 gives

$$S_{1,3,5} = \{1, 3\} \cup [5, 8] \cup [22, 25] \cup [41, 44] \cup \cdots .$$

The proof has two complementary parts. The displayed terms must be safe, meaning that no three distinct earlier displayed terms sum to them; every omitted integer after the seed must be covered by such a triple. Once these two conditions are checked, the greedy construction is forced.

Bosma, Bruin, Fokkink, Grube, Reuijl, and Tromp used Walnut in their study of OEIS problems and conjectured an eventually periodic description of the first regular family. In our notation, their conjecture concerns

$$S_{1,g,g+d} \qquad (2 \leq d \leq g - 1).$$

Conjecture 6 (Bosma–Bruin–Fokkink–Grube–Reuijl–Tromp). *Let $d \geq 2$. For every $g \geq d+1$, the greedy 3-sum-free sequence $S_{1,g,g+d}$, with its prescribed seed $g+d$ included, is characterized by*

$$n \in S_{1,g,g+d}$$

if and only if either

$$n \in \{1, g, 2g + d - 1, 2g + d\},$$

or

$$n \geq g + d \quad \textit{and} \quad n \bmod (5g + 2d) \in \{g + d - 2, g + d - 1, \ldots, 2g + d - 2\}.$$

We prove this conjecture and the adjacent endpoint $d = g$. More generally, after writing $h = y + 1$ and $D = z$, we prove three uniform classification theorems for $S_{1,h-1,D}$: a regular r-band, the boundary point immediately after it, and a dense cap in the following transition window. The regular and boundary theorems provide the stable families; the dense-cap theorem explains the portion of the transition where the terminal packet is large enough for the required interval-sum identities to overlap.

The low-rank cases treated in Section 7 should be read as specializations of this uniform structure. The rank-one regular case is included with a separate direct proof because it lowers the threshold to $g \geq 3$ and resolves the conjecture. The rank-one boundary, rank-two regular, and rank-two boundary cases are concrete forms of the general regular and boundary theorems. They are retained because their interval certificates make the mechanism visible and show the sharp change at the adjacent parameter values.

A Lean 4 verification of the classification statements is available at `https://github.com/Apiros3/greedy-3-sumfree`. The paper gives the conceptual proof structure and selected expanded interval certificates; the repository contains the complete machine-checked details.

The paper is organized as follows. Section 2 fixes the notation and the seeded safety-coverage criterion. Section 3 describes the third-seed parameter geometry. Sections 4 and 5 state the uniform regular, boundary, and dense-cap classifications. Section 6 gives the 3/4 consequence and a general periodic-tail density statement. Section 7 gives the low-rank specializations and endpoint sharpness. Section 8 concludes with the unclassified range and open problems.

2 Notation and Greedy Verification

All intervals in the paper are integer intervals:

$$[u, v] = \{n \in \mathbb{N} : u \leq n \leq v\}.$$

Residues modulo a positive integer M are always taken in $[0, M-1]$; equivalently, $n \bmod M$ denotes the least nonnegative residue. For an integer a and a set A, write

$$a + A = \{a + x : x \in A\}.$$

If $A_1, \ldots, A_k \subset \mathbb{N}$, the restricted sumset

$$(A_1 + \cdots + A_k)_{\text{dist}}$$

is the set of all sums $a_1 + \cdots + a_k$ with $a_i \in A_i$ and with the chosen summands pairwise distinct. When all A_i are the same set H, this is written, for instance, as $(H + H)_{\text{dist}}$ or $(H + H + H)_{\text{dist}}$.

For an integer set Q, write $Q + h = \{q + h : q \in Q\}$. We call Q *h-free* if $Q \cap (Q + h) = \varnothing$. A residue representative is *prefix-only* if it occurs among the finite prefix elements but is absent from every repeated tail block. In a tail-safety argument, if $x_i = n_i M + s_i$, if $\nu = n_1 + n_2 + n_3$, and if $\sigma = s_1 + s_2 + s_3$, then a representation of the target $qM + \rho$ gives the quotient-and-residue equation

$$qM + \rho = \nu M + \sigma.$$

For a proposed candidate set $\mathcal{A}$, an integer $n \in \mathcal{A}$ after the seed is *safe* if no three distinct elements of $\mathcal{A}$ smaller than n sum to n. An omitted integer $n \notin \mathcal{A}$ after the seed is *covered* if such a triple does exist. The next criterion is the verification device used throughout the paper.

Lemma 1 (Seeded greedy criterion)**.** *Let $A \subset \mathbb{N}$ and let $x < y < z$ be positive integers. Suppose that*

$$A \cap [1, z] = \{x, y, z\}.$$

If, for every integer $n > z$,

$$n \in A$$

if and only if there do not exist distinct $a, b, c \in A$, all smaller than n, such that $a+b+c = n$, then $S_{x,y,z} = A$.

Proof. The greedy sequence and A agree up to z by the seed condition. Assume inductively that they agree below $n > z$. The greedy rule accepts n precisely when no triple of distinct earlier greedy terms sums to n. By the induction hypothesis, this is the same as the condition in the lemma statement for A, hence is equivalent to $n \in A$. Induction on n proves the claim. $\square$

Lemma 2 (Distinct sums from an interval)**.** *Let $H = [L, U]$ and let $k \geq 1$. If $U - L + 1 \geq k$, then the sums of k distinct elements of H are exactly*

$$\left[kL + \binom{k}{2},\, kU - \binom{k}{2}\right].$$

Proof. The minimum is $L + (L+1) + \cdots + (L+k-1)$, and the maximum is $(U-k+1) + \cdots + U$. Let $m = U - L$ and $C = m - k + 1$. It remains to show that every integer T with $0 \leq T \leq kC$ can be added to the minimum sum while preserving distinctness and the upper bound. Write

$$T = qk + s, \qquad 0 \leq s < k.$$

If $s = 0$, take $u_1 = \cdots = u_k = q$. If $s > 0$, take

$$u_1 = \cdots = u_{k-s} = q, \qquad u_{k-s+1} = \cdots = u_k = q + 1.$$

In either case $0 \leq u_1 \leq \cdots \leq u_k \leq C$ and $\sum_i u_i = T$. The k numbers

$$L + u_1, \quad L + 1 + u_2, \quad \ldots, \quad L + k - 1 + u_k$$

are distinct elements of $[L, U]$ with sum

$$kL + \binom{k}{2} + T.$$

This realizes every sum between the minimum and maximum. $\square$

Lemma 3 (Interval-chain criterion)**.** *Let $[L_1, U_1], \ldots, [L_m, U_m]$ be integer intervals ordered by their left endpoints. If $L_{j+1} \leq U_j + 1$ for every $j < m$, then their union is the integer interval $[L_1, \max_j U_j]$.*

Proof. The hypothesis says that each interval begins before or immediately after the preceding covered range ends. Induction on m gives the claim. $\square$

3 Geometry of the Third Seed

It is useful to normalize the third seed by writing

$$h = y + 1, \qquad D = z.$$

For $S_{1,h-1,D}$, we call r the *rank* of the recurrent block

$$(2r-1)h + 1 \le D \le (2r+1)h.$$

The same regular-boundary-transition geometry recurs as r increases. Within a rank-r block, the regimes are:

Regime	Values of D	Number of values
Regular band	$(2r-1)h+1 \le D \le 2rh-2$	$h-2$
Boundary point	$D = 2rh-1$	1
Transition window	$D = 2rh+t,\ 0 \le t \le h$	$h+1$
Dense cap classified here	$\lceil (h+2)/2 \rceil \le t \le h-2$	$h-1-\lceil (h+2)/2 \rceil$

Equivalently, the regular bands occupy

$$(2r-1)h + 1 \le D \le 2rh - 2,$$

their boundary point is $D = 2rh - 1$, and the transition window following that boundary is

$$D = 2rh + t, \qquad 0 \le t \le h.$$

The dense-cap theorem below classifies a dense upper portion of this transition window. Specifically, Theorem 3 applies to

$$D = 2rh + t, \qquad \left\lceil \frac{h+2}{2} \right\rceil \le t \le h-2.$$

The remaining transition parameters are the sparse lower portion $0 \le t < \lceil (h+2)/2 \rceil$ and the two upper endpoints $t = h-1, h$.

4 Regular and Boundary Classifications

Theorem 1 (Regular r-bands)**.** *Let $h \ge 6$, $r \ge 1$, and*

$$(2r-1)h + 1 \le D \le 2rh - 2.$$

For $0 \le i < r$, put

$$H_i = [D + 2ih, D + 2ih + h - 1], \qquad I_i = [D + 2ih - 2, D + 2ih + h - 3],$$

and

$$M = 2D + (6r-3)h - 3.$$

Then

$$S_{1,h-1,D} = \{1, h-1\} \cup \bigcup_{i=0}^{r-1} H_i \cup \bigcup_{q \ge 1} \bigcup_{i=0}^{r-1} (qM + I_i).$$

Proof sketch. Let $\mathcal{A}$ be the candidate set in the theorem statement. We verify the two conditions of Lemma 1.

For prefix safety, separate triples by how many summands come from the intervals H_i. A triple using both small seeds sends a prefix element u to $u+h$, which lies in the gap after the interval containing u. Triples using at least two interval elements are excluded from the prefix intervals by the lower endpoint bounds, and triples using an earlier H_j cannot land inside a later H_i except through the same h-shift gap.

For tail safety, consider a putative representation $qM + \rho = x_1 + x_2 + x_3$, with $q \geq 1$ and $\rho \in \bigcup_i I_i$. No summand from the current tail block with residue at least ρ is smaller than the target. Reducing the remaining possibilities modulo M, the three-residue sum σ lies in a bounded range, so $\sigma - \rho$ can have only the listed carry values. The only residue sums congruent to an occupied tail residue use prefix-only endpoint residues. In those cases the quotient contribution is $\nu = 0$, contradicting the equation $qM + \rho = \nu M + \sigma$ with $q \geq 1$.

For coverage, the omitted prefix and tail gaps are filled by chains of restricted interval sums from Lemma 2. The regular-band inequalities are precisely the hypotheses that make successive intervals adjacent or overlapping, as in Lemma 3. The complete endpoint and residue checks are contained in the linked Lean verification. □

Theorem 2 (Rank-r boundary points)**.** *Let $r \geq 1$, $h \geq 6$, and $D = 2rh - 1$. For $0 \leq i < r$, put*

$$H_i = [D + 2ih, D + 2ih + h - 1], \qquad K_i = [D + 2ih - 1, D + 2ih + h - 2].$$

Let

$$c = 4rh - 1, \qquad M = (10r - 2)h - 3.$$

Then

$$S_{1,h-1,D} = \{1, h-1, c\} \cup \bigcup_{i=0}^{r-1} H_i \cup \bigcup_{q \geq 1} \bigcup_{i=0}^{r-1} (qM + K_i).$$

Proof sketch. The boundary point is the first value after the regular band. The regular tail residues would no longer cover all prefix-safe elements: the regular-band candidate omits the first new safe prefix element

$$c = 4rh - 1$$

before periodicity resumes. It is safe because it lies after the last sum obtained by a low-seed shift but before the first possible sum using two large prefix elements. Including c shifts the recurrent residue intervals from I_i to K_i and changes the eventual period to the displayed value of M.

With this singleton included, the proof again follows Lemma 1. Prefix safety comes from the separation of the H_i's and from the placement of c. Tail safety is the same quotient-and-residue argument as in the regular case, with c treated as a prefix-only residue. Coverage is a finite collection of restricted interval-sum chains whose consecutive overlaps are exactly the boundary endpoint checks. The complete endpoint and residue checks are contained in the linked Lean verification. □

5 Dense Transition Caps

Theorem 3 (Dense-cap transition)**.** *Let $h \geq 6$, $r \geq 1$, and*

$$\left\lceil \frac{h+2}{2} \right\rceil \leq t \leq h - 2.$$

Put

$$D = 2rh + t, \qquad M = 10D + 3.$$

For $s \in \{0,1,2\}$, define

$$Q_s = \bigcup_{i=0}^{r-1} [2ih, 2ih + h - 1] \cup [2rh, 2rh + t + 1 - s].$$

Let

$$E = D + Q_0, \qquad X = D + Q_1, \qquad W = D + Q_2,$$
$$F = 5D + 1 + X, \qquad Y = 5D + 2 + W.$$

Then

$$S_{1,h-1,D} = \{1, h-1\} \cup E \cup F \cup \bigcup_{q \geq 1} (qM + X) \cup \bigcup_{q \geq 1} (qM + Y).$$

Membership is eventually periodic modulo $M = 10D + 3$, and the occupied tail residues are $X \cup Y$.

Here E and F are the two finite prefix packets beyond the small seeds. The sets X and Y are the corresponding shortened low and high tail-residue packets; the auxiliary packet W records the two-unit shortening needed to define Y. The dense condition $2t \geq h + 2$ is an overlap condition: it says that the terminal caps in the packet-sum chains below are long enough to close the final possible gap.

Lemma 4 (Dense tail-residue classification)**.** *Assume the notation of Theorem 3, and put*

$$\Sigma = \{1, h-1\} \cup E \cup F \cup X \cup Y.$$

If three pairwise distinct representatives from Σ have sum congruent modulo M to a residue $\rho \in X \cup Y$, then the only possible residue category is

$$\{1, h-1, f\}, \qquad f \in F \setminus Y, \qquad f + h \in Y.$$

The remaining endpoint possibilities require one of the prefix-only residues $2D + 1$ or $6D + 1$ to be used twice.

Proof. The low packet residues lie in $[D, 2D + 1]$, the high packet residues in $[6D + 1, 7D + 1]$, and the repeated tail residues are

$$X \subset [D, 2D], \qquad Y \subset [6D + 2, 7D + 1].$$

The case split is by the number of small residues and by whether packet residues are low or high.

Residue category	Possible target behavior	Exclusion or survivor
$\{1, h-1, u\}$, u low	would give $u + h$	h-freeness of the low packets puts $u + h$ in a gap.
$\{1, h-1, f\}$, f high	would give $f + h$	survives exactly when $f \in F \setminus Y$ and $f + h \in Y$.
low packet residues, with at most one small residue	can reach Y only at the top endpoint	the endpoint calculation forces the prefix-only residue $2D + 1$ twice.
one high packet residue without both small residues	falls between the occupied low and high tail ranges	the only endpoint alternatives again force a repeated prefix-only endpoint.
two or three high packet residues	after subtracting one carry of M, can reach X only at the bottom endpoint	the endpoint calculation forces the prefix-only residue $6D + 1$ twice.

These alternatives exhaust the residue categories. The endpoint calculations are finite interval comparisons; the linked Lean verification contains the complete checked inequalities. The table leaves only the stated category. □

Lemma 5 (Dense packet sums). *Under the hypotheses of Theorem 3, the packet sums satisfy*

$$(Q_0+Q_0)_{\mathrm{dist}} \cup \big(h-2+(Q_0+Q_0)_{\mathrm{dist}}\big) = [1, 2D+h-1],$$

$$(Q_a+Q_b) \cup (h-2+Q_a+Q_b) = [0, 2D+h-a-b] \qquad (a,b \in \{0,1\}),$$

$$(Q_0+Q_0+Q_0)_{\mathrm{dist}} = [3, 3D],$$

and

$$(Q_0+Q_0)_{\mathrm{dist}} + Q_s = [1, 3D+2-s] \qquad (s \in \{0,1,2\}).$$

Proof. When two summands come from disjoint packet intervals, distinctness is automatic; the subscript dist is needed only when two or more summands may come from the same packet. Each fixed choice of packet cells gives an integer interval by Lemma 2. Ordering those output intervals by their lower endpoints, all consecutive pairs overlap or are adjacent by the interval-chain criterion (Lemma 3) except possibly at the final terminal cap. The inequality $2t \geq h+2$ is exactly the remaining endpoint inequality for that terminal overlap. This proves the displayed packet identities. □

Proof of Theorem 3. Let A be the candidate set in the theorem statement. The proof is another application of Lemma 1: prove safety for candidate elements and coverage for every later omitted integer.

First consider prefix safety. Each Q_s is h-free: adding h to any point of Q_s lands in a gap between adjacent packet intervals or beyond the terminal cap. Hence a triple using the two small seeds has the form $1+(h-1)+u = u+h$ and misses the packet containing u. This proves safety inside E. For F, triples from E are at most $6D$, while $\min F = 6D+1$; triples using an earlier element of F and an element of E exceed $\max F = 7D+1$; and the same h-free argument excludes the small-seed shift inside F.

For tail safety, reduce a putative representation of $qM+\rho$, with $q \geq 1$ and $\rho \in X \cup Y$, modulo M. By Lemma 4, the only possible residue category is $\{1, h-1, f\}$ with $f \in F \setminus Y$ and $f+h \in Y$. In this surviving category all three summands are prefix elements, so their quotient contribution is $\nu = 0$. The quotient-and-residue equation

$$qM + \rho = \nu M + \sigma$$

then contradicts $q \geq 1$. Thus all tail candidates are safe.

It remains to cover the omitted integers. Internal gaps inside E, F, $qM+X$, and $qM+Y$ are covered by adding $1+(h-1)$. The remaining gaps are covered by the following families; the endpoint identities are Lemma 5.

Target gap	Covering families	Distinctness and earlier-term check
$[2D+2, 6D]$	$1+(E+E)_{\text{dist}}$, $(h-1)+(E+E)_{\text{dist}}$, $(E+E+E)_{\text{dist}}$	restricted sums provide distinct prefix terms.
$[7D+2, 11D+2]$	$1+E+F$, $(h-1)+E+F$, $(E+E)_{\text{dist}}+F$	E and F are disjoint; repeated E-choices are restricted.
low-to-high periodic gap	$1+E+X$, $(h-1)+E+X$, $X+(E+E)_{\text{dist}}$	the X-summand is in the current block with lower residue than the target.
high-to-next-low periodic gap	$1+F+X$, $(h-1)+F+X$, $Y+(E+E)_{\text{dist}}$	the periodic summand is either from the same block with lower residue or from the preceding high block.

Thus every omitted integer after the seed is represented by three distinct smaller elements of A. Safety and coverage prove the claimed greedy sequence by Lemma 1. □

6 Consequences

Corollary 1 (Asymptotic coverage of third-seed positions)**.** *Fix $h \geq 6$. In each recurrent block*

$$(2r-1)h+1 \leq D \leq (2r+1)h$$

of $2h$ possible third-seed positions for $S_{1,h-1,D}$, the regular band, boundary point, and dense-cap transition classify

$$2h-2-\left\lceil \frac{h+2}{2} \right\rceil$$

positions. Consequently the classified proportion tends to $3/4$ as $h \to \infty$.

Proof. The regular band contributes $h-2$ positions, the boundary theorem one position, and the dense-cap theorem

$$h-1-\left\lceil \frac{h+2}{2} \right\rceil$$

positions. This is the count in the corollary, and division by $2h$ gives the claimed limit. □

Proposition 1 (Density and generating function of a periodic tail)**.** *Let $M \geq 1$, let $P \subset \mathbb{N}$ be finite, and let $R \subseteq [0, M-1]$. Suppose*

$$A = P \cup \bigcup_{q \geq 1} (qM + R),$$

where P contains the complete preperiod. Then the natural density

$$\lim_{N \to \infty} \frac{|A \cap [1, N]|}{N}$$

exists and is

$$\frac{|R|}{M}.$$

Moreover the ordinary generating function is rational:

$$F_A(x) := \sum_{a \in A} x^a = \sum_{p \in P} x^p + \frac{x^M \sum_{\rho \in R} x^\rho}{1 - x^M}.$$

Proof. The density statement follows by counting one complete residue pattern in each period. The generating-function formula is the sum of a finite polynomial and finitely many geometric series, one for each occupied tail residue. □

7 Low-Rank Specializations and Sharpness

This section records the rank-one and rank-two specializations of the regular and boundary classifications. The rank-one regular theorem is proved directly because the direct argument lowers the general threshold from $g \geq 5$ to $g \geq 3$ and resolves [1, Conjecture 6]. The other three statements follow immediately from Theorems 1 and 2; expanded direct certificates are included to display the interval-covering mechanism and the change at the adjacent boundary values.

The following table records how the explicit families in this section fit into the general theorems.

Explicit result	General source	Specialization
Theorem 4	Theorem 1	$r = 1,\ h = g + 1,\ D = g + d$; direct proof lowers $g \geq 5$ to $g \geq 3$.
Corollary 3	Theorem 2	$r = 1,\ h = g + 1,\ D = 2g + 1$.
Corollary 4	Theorem 1	$r = 2,\ h = g + 1,\ D = g + d$.
Corollary 5	Theorem 2	$r = 2,\ h = g + 1,\ D = 4g + 3$.

7.1 Rank One: Regular Band

Theorem 4 (Rank-one regular band)**.** *Let $g \geq 3$, $2 \leq d \leq g$, and put*

$$M = 5g + 2d, \qquad a = g + d - 2, \qquad b = 2g + d - 2.$$

Let

$$P = \{1, g\} \cup [g + d, 2g + d], \qquad B_q = [qM + a, qM + b] \quad (q \geq 1).$$

Then

$$S_{1,g,g+d} = P \cup B_1 \cup B_2 \cup \cdots .$$

Equivalently, an integer n belongs to $S_{1,g,g+d}$ if and only if

$$n \in \{1, g, 2g + d - 1, 2g + d\}$$

or

$$n \geq g + d \quad \textit{and} \quad n \bmod M \in [g + d - 2, 2g + d - 2].$$

Proof. Let

$$A = P \cup B_1 \cup B_2 \cup \cdots .$$

By Lemma 1, it is enough to prove that every element of A after the seed $g + d$ is safe, and that every later integer not in A is a sum of three distinct smaller elements of A.

Prefix safety. First let $n \in [g + d + 1, 2g + d]$. Before n is reached, the smallest possible sum of three distinct earlier terms is

$$1 + g + (g + d) = 2g + d + 1,$$

so these prefix elements are safe.

Periodic-block safety. Now let $n = qM+r$, where $q \geq 1$ and $a \leq r \leq b$. Suppose, for a contradiction, that $n = x_1 + x_2 + x_3$ for distinct smaller $x_i \in A$. No summand can lie in B_q, since any such triple is at least

$$(qM + a) + 1 + g = qM + 2g + d - 1 > qM + b.$$

Thus all summands lie in the prefix or in earlier periodic blocks. Let s_i be the residue of x_i modulo M, chosen in $[0, M-1]$, and set $\sigma = s_1 + s_2 + s_3$. The smallest possible residue sum of three distinct elements of A is at least

$$1 + g + a = 2g + d - 1.$$

For the upper bound, the only residues above b are

$$b + 1 = 2g + d - 1, \qquad b + 2 = 2g + d,$$

and each occurs only once, in the prefix. Every other residue is at most b. Hence

$$\sigma \leq (b + 2) + (b + 1) + b = 6g + 3d - 3.$$

Since $r \in [a, b]$,

$$1 \leq \sigma - r \leq (6g + 3d - 3) - (g + d - 2) = 5g + 2d - 1 = M - 1.$$

But $qM + r = x_1 + x_2 + x_3$ forces $\sigma - r$ to be a multiple of M, a contradiction. Therefore all periodic-block elements are safe.

Initial gap coverage. It remains to cover the gaps. Write

$$H = [g + d, 2g + d].$$

The only prefix gap before B_1 is

$$[2g + d + 1, M + a - 1] = [2g + d + 1, 6g + 3d - 3].$$

By Lemma 2,

$$(H + H)_{\text{dist}} = [2g + 2d + 1, 4g + 2d - 1],$$

and

$$(H + H + H)_{\text{dist}} = [3g + 3d + 3, 6g + 3d - 3].$$

The prefix gap is covered by

$$1 + g + H = [2g + d + 1, 3g + d + 1],$$

$$1 + (H + H)_{\text{dist}} = [2g + 2d + 2, 4g + 2d],$$

$$g + (H + H)_{\text{dist}} = [3g + 2d + 1, 5g + 2d - 1],$$

$$(H + H + H)_{\text{dist}} = [3g + 3d + 3, 6g + 3d - 3].$$

These intervals overlap consecutively because

$$2g + 2d + 2 \leq 3g + d + 2, \qquad 3g + 2d + 1 \leq 4g + 2d + 1,$$

and

$$3g + 3d + 3 \leq 5g + 2d.$$

The first and third inequalities use $d \le g$, with the third also using $g \ge 3$.

Periodic gap coverage. Fix $q \ge 1$. The gap after B_q and before B_{q+1} is

$$[qM + b + 1, (q+1)M + a - 1] = qM + [2g + d - 1, 6g + 3d - 3].$$

The following subsets of $(P + P)_{\text{dist}}$ suffice:

$$1 + g = g + 1, \qquad 1 + H = [g + d + 1, 2g + d + 1],$$

$$g + H = [2g + d, 3g + d], \qquad (H + H)_{\text{dist}} = [2g + 2d + 1, 4g + 2d - 1].$$

Adding the residue interval $[a, b]$ of B_q gives

$$[a, b] + (1 + g) = [2g + d - 1, 3g + d - 1],$$

$$[a, b] + (1 + H) = [2g + 2d - 1, 4g + 2d - 1],$$

$$[a, b] + (g + H) = [3g + 2d - 2, 5g + 2d - 2],$$

$$[a, b] + (H + H)_{\text{dist}} = [3g + 3d - 1, 6g + 3d - 3].$$

These intervals cover the whole offset range because

$$2g + 2d - 1 \le 3g + d, \qquad 3g + 2d - 2 \le 4g + 2d,$$

and

$$3g + 3d - 1 \le 5g + 2d - 1.$$

Thus every element of the periodic gap has the form

$$(qM + t) + p_1 + p_2,$$

with $t \in [a, b]$ and distinct $p_1, p_2 \in P$. The periodic summand is smaller than the target because $t \le b$, while the target offset is at least $b + 1$. Hence all gap elements are covered. $\square$

Corollary 2. *For $g \ge 3$ and $2 \le d \le g - 1$, the formula in Theorem 4 resolves [1, Conjecture 6].*

7.2 Rank One: Boundary Point

Corollary 3 (Rank-one boundary point)**.** *Let $g \ge 5$, put $M = 8g + 5$, and define*

$$H = [2g + 1, 3g + 1], \qquad c = 4g + 3.$$

Let

$$P = \{1, g, c\} \cup H, \qquad B_q = [qM + 2g, qM + 3g] \quad (q \ge 1).$$

Then

$$S_{1,g,2g+1} = P \cup B_1 \cup B_2 \cup \cdots .$$

Proof. Substituting $r = 1$, $h = g+1$, and $D = 2g+1$ in Theorem 2 gives exactly the stated formula. For comparison with the rank-one regular proof, we include an expanded direct certificate.

Let $A = P \cup B_1 \cup B_2 \cup \cdots$. We apply Lemma 1.

Prefix safety. Every $n \in [2g+2, 3g+1]$ is safe because

$$1 + g + (2g+1) = 3g + 2 > n.$$

The singleton $c = 4g+3$ is safe as well. If a triple from $\{1, g\} \cup H$ uses at most one element of H, then it is at most

$$1 + g + (3g+1) = 4g + 2,$$

while if it uses at least two elements of H, then it is at least

$$1 + (2g+1) + (2g+2) = 4g + 4.$$

Thus no earlier triple sums to $4g+3$.

Periodic-block safety. Now let $n = qM + r$, where $q \geq 1$ and $2g \leq r \leq 3g$. No summand in a representation of n can lie in B_q, since

$$(qM + 2g) + 1 + g = qM + 3g + 1 > qM + 3g.$$

Therefore all summands would have to lie in the prefix or earlier blocks. Their residues belong to

$$\{1, g, 4g+3\} \cup [2g, 3g+1],$$

so their residue sum σ satisfies

$$3g + 1 \leq \sigma \leq (4g+3) + (3g+1) + 3g = 10g + 4.$$

Since $r \in [2g, 3g]$,

$$1 \leq \sigma - r \leq 8g + 4 = M - 1,$$

contradicting the required congruence. Hence all periodic elements are safe.

Prefix gap coverage. The gap between H and c is

$$[3g+2, 4g+2] = 1 + g + H.$$

The next prefix gap is

$$[4g+4, M+2g-1] = [4g+4, 10g+4].$$

By Lemma 2,

$$(H+H)_{\text{dist}} = [4g+3, 6g+1], \qquad (H+H+H)_{\text{dist}} = [6g+6, 9g].$$

Thus this gap is covered by

$$1 + (H+H)_{\text{dist}} = [4g+4, 6g+2],$$

$$g + (H+H)_{\text{dist}} = [5g+3, 7g+1],$$

$$(H+H+H)_{\text{dist}} = [6g+6, 9g],$$

$$c + (H+H)_{\text{dist}} = [8g+6, 10g+4].$$

The consecutive overlaps are

$$5g+3 \le 6g+3, \qquad 6g+6 \le 7g+2, \qquad 8g+6 \le 9g+1,$$

which hold for $g \ge 5$.

Periodic gap coverage. Finally fix $q \ge 1$. The gap after B_q is

$$[qM+3g+1,(q+1)M+2g-1] = qM+[3g+1,10g+4].$$

Using one element from B_q and two distinct prefix elements gives

$$[2g,3g]+(1+g) = [3g+1,4g+1],$$

$$[2g,3g]+(1+H) = [4g+2,6g+2],$$
$$[2g,3g]+(H+H)_{\text{dist}} = [6g+3,9g+1],$$
$$[2g,3g]+(H+c) = [8g+4,10g+4].$$

These cover the entire offset interval: the first three are adjacent, and the last overlaps the third because $8g+4 \le 9g+2$. The block term is smaller than the target since its offset is at most $3g$, while every target offset in the gap is at least $3g+1$. This covers all excluded integers. □

7.3 Rank Two: Regular Band

Corollary 4 (Rank-two regular band)**.** *Let $g \ge 5$, $2g+4 \le d \le 3g+2$, and put*

$$M = 11g+2d+6.$$

Define

$$H = [g+d,2g+d], \qquad C = [3g+d+2,4g+d+2],$$

and

$$I = [g+d-2,2g+d-2], \qquad J = [3g+d,4g+d].$$

Then

$$S_{1,g,g+d} = \{1,g\} \cup H \cup C \cup \bigcup_{q\ge1}(qM+I) \cup \bigcup_{q\ge1}(qM+J).$$

Proof. Substituting $r=2$, $h=g+1$, and $D=g+d$ in Theorem 1 gives exactly the stated formula. We include an expanded direct certificate for the interval endpoints.

Let A be the candidate set in the corollary statement and put $P=\{1,g\}\cup H\cup C$.

Prefix safety. The elements of H after $g+d$ are safe because they are smaller than

$$1+g+(g+d) = 2g+d+1.$$

The elements of C are also safe. A triple from $\{1,g\}\cup H$ of the form $1+g+h$ is at most $3g+d+1$, while any triple using at least two elements of H is at least

$$1+(g+d)+(g+d+1) = 2g+2d+2 > 4g+d+2,$$

since $d \ge 2g+4$. A triple using an earlier element of C is at least $1+g+(3g+d+2) = 4g+d+3$, above all of C.

Periodic-block safety. Now let $n = qM + r$, where $q \geq 1$ and $r \in I \cup J$, and suppose that $n = x_1 + x_2 + x_3$ for distinct smaller elements of A. Write $x_i = n_i M + s_i$, with $n_i = 0$ for prefix elements and with $s_i \in I \cup J$ for periodic elements. Let

$$\nu = n_1 + n_2 + n_3, \qquad \sigma = s_1 + s_2 + s_3.$$

The available residues lie in

$$\{1, g\} \cup [g + d - 2, 2g + d] \cup [3g + d, 4g + d + 2].$$

Therefore

$$2g + d - 1 \leq \sigma \leq (4g + d + 2) + (4g + d + 1) + (4g + d) = 12g + 3d + 3.$$

Since $r \in I \cup J$, we have $-M < \sigma - r < M$. The equality $qM + r = \nu M + \sigma$ therefore forces $\sigma = r$ and $\nu = q$. This is impossible when $r \in I$, because $\sigma \geq 2g + d - 1 > 2g + d - 2$. When $r \in J$, the only way to have a residue sum in J is

$$\sigma = 1 + g + \tau, \qquad \tau \in \{2g + d - 1, 2g + d\}.$$

Indeed, one high residue is already too large with $1 + g$, and two residues at least $g + d - 2$ sum with 1 to more than $4g + d$. But the residues $2g + d - 1$ and $2g + d$ occur only in the prefix interval H, so such a triple has $\nu = 0$, contradicting $\nu = q \geq 1$. Thus all periodic elements are safe.

Prefix gap coverage. The gap between H and C is

$$[2g + d + 1, 3g + d + 1] = 1 + g + H.$$

The gap between C and the first periodic interval is

$$[4g + d + 3, M + g + d - 3] = [4g + d + 3, 12g + 3d + 3].$$

Using Lemma 2, this gap is covered by

$$\begin{aligned}
1 + g + C &= [4g + d + 3, 5g + d + 3], \\
1 + (H + H)_{\text{dist}} &= [2g + 2d + 2, 4g + 2d], \\
g + (H + H)_{\text{dist}} &= [3g + 2d + 1, 5g + 2d - 1], \\
1 + H + C &= [4g + 2d + 3, 6g + 2d + 3], \\
g + H + C &= [5g + 2d + 2, 7g + 2d + 2], \\
(H + H + H)_{\text{dist}} &= [3g + 3d + 3, 6g + 3d - 3], \\
(H + H)_{\text{dist}} + C &= [5g + 3d + 3, 8g + 3d + 1], \\
H + (C + C)_{\text{dist}} &= [7g + 3d + 5, 10g + 3d + 3], \\
(C + C + C)_{\text{dist}} &= [9g + 3d + 9, 12g + 3d + 3].
\end{aligned}$$

For integer intervals, consecutive coverage is checked by requiring each next lower endpoint to be at most the previous upper endpoint plus 1. The required inequalities are

$$\begin{aligned}
2g + 2d + 2 &\le 5g + d + 4 && (d \le 3g + 2),\\
3g + 2d + 1 &\le 4g + 2d + 1, &&\\
4g + 2d + 3 &\le 5g + 2d && (g \ge 3),\\
5g + 2d + 2 &\le 6g + 2d + 4, &&\\
3g + 3d + 3 &\le 7g + 2d + 3 && (d \le 4g),\\
5g + 3d + 3 &\le 6g + 3d - 2 && (g \ge 5),\\
7g + 3d + 5 &\le 8g + 3d + 2 && (g \ge 3),\\
9g + 3d + 9 &\le 10g + 3d + 4 && (g \ge 5).
\end{aligned}$$

Thus the whole prefix gap is covered.

Periodic gap coverage. Fix $q \ge 1$. The gap between $qM + I$ and $qM + J$ is

$$qM + [2g + d - 1, 3g + d - 1] = qM + I + (1 + g).$$

The gap after $qM + J$ is

$$qM + [4g + d + 1, 12g + 3d + 3].$$

It is covered by one element from $qM + I$ or $qM + J$, together with two distinct prefix elements, through

$$\begin{aligned}
J + (1 + g) &= [4g + d + 1, 5g + d + 1],\\
I + (1 + H) &= [2g + 2d - 1, 4g + 2d - 1],\\
I + (g + H) &= [3g + 2d - 2, 5g + 2d - 2],\\
I + (1 + C) &= [4g + 2d + 1, 6g + 2d + 1],\\
I + (g + C) &= [5g + 2d, 7g + 2d],\\
I + (H + H)_{\mathrm{dist}} &= [3g + 3d - 1, 6g + 3d - 3],\\
I + (H + C) &= [5g + 3d, 8g + 3d],\\
J + (H + C) &= [7g + 3d + 2, 10g + 3d + 2],\\
J + (C + C)_{\mathrm{dist}} &= [9g + 3d + 5, 12g + 3d + 3].
\end{aligned}$$

The consecutive overlaps are

$$\begin{aligned}
2g + 2d - 1 &\le 5g + d + 2 && (d \le 3g + 3),\\
3g + 2d - 2 &\le 4g + 2d, &&\\
4g + 2d + 1 &\le 5g + 2d - 1 && (g \ge 2),\\
5g + 2d &\le 6g + 2d + 2, &&\\
3g + 3d - 1 &\le 7g + 2d + 1 && (d \le 4g + 2),\\
5g + 3d &\le 6g + 3d - 2 && (g \ge 2),\\
7g + 3d + 2 &\le 8g + 3d + 1, &&\\
9g + 3d + 5 &\le 10g + 3d + 3 && (g \ge 2).
\end{aligned}$$

The periodic summand is smaller than the target because its offset is at most $4g + d$, while every target offset in this gap is at least $4g + d + 1$. This covers all excluded integers, and Lemma 1 proves the theorem. $\square$

The formula in Corollary 4 is sharp at its adjacent d-boundaries. At the lower neighbor $d = 2g+3$, the candidate element $3M+4g+d$ in the J-block predicted by formally extending Corollary 4 is not safe:

$$3M + 4g + d = 1 + (M + g + d - 2) + (2M + g + d - 2).$$

At the upper neighbor $d = 3g + 3$, the same regular formula omits the safe prefix-gap element

$$2g + 2d + 1 = 8g + 7.$$

The boundary theorem below inserts exactly this forced singleton and changes the eventual period. These endpoint checks show the sharpness of the regular formula, not a failure of eventual periodicity at those parameters.

7.4 Rank Two: Boundary Point

Corollary 5 (Rank-two boundary point)**.** *Let $g \geq 5$, put $M = 18g + 15$, and define*

$$H = [4g + 3, 5g + 3], \qquad C = [6g + 5, 7g + 5], \qquad c = 8g + 7,$$

$$I = [4g + 2, 5g + 2], \qquad J = [6g + 4, 7g + 4].$$

Then

$$S_{1,g,4g+3} = \{1, g, c\} \cup H \cup C \cup \bigcup_{q\geq 1}(qM + I) \cup \bigcup_{q\geq 1}(qM + J).$$

Proof. Substituting $r = 2$, $h = g+1$, and $D = 4g+3$ in Theorem 2 gives exactly the stated formula. We include an expanded direct certificate to show how the forced singleton enters the safety and coverage checks.

Let A be the candidate set in the corollary statement.

Prefix safety. The elements of H after the seed $4g + 3$ are safe because they are smaller than

$$1 + g + (4g + 3) = 5g + 4.$$

The interval C is safe: $1 + g + H = [5g + 4, 6g + 4]$ ends before C, the interval $1 + (H + H)_{\text{dist}}$ begins at $8g + 8$, and a triple using an earlier element of C is at least $1 + g + (6g + 5) = 7g + 6$, above all of C. The singleton $c = 8g + 7$ is also safe. Before c, triples $1 + g + C$ fill $[7g + 6, 8g + 6]$; triples with no C either end at $6g + 4$ or begin at $8g + 8$; and all other triples using C begin at $1 + H + C = 10g + 9$.

Periodic-block safety. Now let $n = qM + r$, where $q \geq 1$ and $r \in I \cup J$, and suppose

$$n = x_1 + x_2 + x_3$$

for distinct smaller elements of A. Write $x_i = n_iM + s_i$, with prefix elements having $n_i = 0$. The residues satisfy

$$s_i \in \mathcal{R} = \{1, g, 8g + 7\} \cup [4g + 2, 5g + 3] \cup [6g + 4, 7g + 5].$$

If $\nu = n_1 + n_2 + n_3$ and $\sigma = s_1 + s_2 + s_3$, then

$$5g + 3 \leq \sigma \leq 22g + 16.$$

The lower bound is $1 + g + (4g + 2)$. For the upper bound, the only residues above $7g + 4$ are the prefix-only residues $7g + 5$ and $8g + 7$, so the maximum distinct residue sum is

$$(8g + 7) + (7g + 5) + (7g + 4) = 22g + 16.$$

Since $r \in [4g+2, 7g+4]$, we have $-M < \sigma - r < M$. The equality $qM + r = \nu M + \sigma$ therefore forces $\sigma = r$ and $\nu = q$. This is impossible for $r \in I$, because $\sigma \geq 5g+3 > 5g+2$. For $r \in J$, any residue sum at most $7g+4$ must use 1 and g, since

$$1 + (4g+2) + (4g+2) = 8g+5 > 7g+4.$$

Thus a sum landing in J has the form $1 + g + \tau$ with $\tau \in [4g+2, 5g+3]$. The only such value in J is

$$1 + g + (5g+3) = 6g+4.$$

All three residues in this representation are prefix-only, so $\nu = 0$, contrary to $\nu = q \geq 1$. Hence all periodic elements are safe.

Prefix gap coverage. The first two prefix gaps are

$$[5g+4, 6g+4] = 1 + g + H, \qquad [7g+6, 8g+6] = 1 + g + C.$$

The remaining prefix gap $[8g+8, M+4g+1] = [8g+8, 22g+16]$ is covered by

$$\begin{aligned}
1 + (H+H)_{\text{dist}} &= [8g+8, 10g+6], \\
g + (H+H)_{\text{dist}} &= [9g+7, 11g+5], \\
1 + H + C &= [10g+9, 12g+9], \\
g + H + C &= [11g+8, 13g+8], \\
(H+H+H)_{\text{dist}} &= [12g+12, 15g+6], \\
(H+H)_{\text{dist}} + C &= [14g+12, 17g+10], \\
H + (C+C)_{\text{dist}} &= [16g+14, 19g+12], \\
(C+C+C)_{\text{dist}} &= [18g+18, 21g+12], \\
c + (C+C)_{\text{dist}} &= [20g+18, 22g+16].
\end{aligned}$$

The consecutive-overlap checks are

$$\begin{aligned}
9g+7 &\leq 10g+7, \\
10g+9 &\leq 11g+6 \qquad (g \geq 3), \\
11g+8 &\leq 12g+10, \\
12g+12 &\leq 13g+9 \qquad (g \geq 3), \\
14g+12 &\leq 15g+7 \qquad (g \geq 5), \\
16g+14 &\leq 17g+11 \qquad (g \geq 3), \\
18g+18 &\leq 19g+13 \qquad (g \geq 5), \\
20g+18 &\leq 21g+13 \qquad (g \geq 5).
\end{aligned}$$

Hence the prefix gaps are covered.

Periodic gap coverage. Fix $q \geq 1$. The internal periodic gap is

$$qM + [5g+3, 6g+3] = qM + I + (1+g).$$

The post-J gap is $qM + [7g+5, 22g+16]$. It is covered by

$$\begin{aligned}
J + (1+g) &= [7g+5, 8g+5],\\
I + (1+H) &= [8g+6, 10g+6],\\
I + (g+H) &= [9g+5, 11g+5],\\
I + (1+C) &= [10g+8, 12g+8],\\
I + (H+H)_{\text{dist}} &= [12g+9, 15g+7],\\
I + (H+C) &= [14g+10, 17g+10],\\
J + (H+C) &= [16g+12, 19g+12],\\
J + (C+C)_{\text{dist}} &= [18g+15, 21g+13],\\
J + (C+c) &= [20g+16, 22g+16].
\end{aligned}$$

These intervals overlap because

$$\begin{aligned}
&8g+6 = (8g+5)+1,\\
&9g+5 \le 10g+7,\\
&10g+8 \le 11g+6 &&(g \ge 2),\\
&12g+9 = (12g+8)+1,\\
&14g+10 \le 15g+8 &&(g \ge 2),\\
&16g+12 \le 17g+11,\\
&18g+15 \le 19g+13 &&(g \ge 2),\\
&20g+16 \le 21g+14 &&(g \ge 2).
\end{aligned}$$

In each periodic-gap representation the periodic summand has offset below the target offset: at most $5g+2$ in the internal gap and at most $7g+4$ in the post-J gap. Thus all excluded integers are covered, and Lemma 1 completes the proof. □

For the rank-two boundary family of Corollary 5, membership is eventually periodic modulo $M = 18g+15$, with occupied tail residues

$$I = [4g+2, 5g+2], \qquad J = [6g+4, 7g+4].$$

Therefore its natural density is

$$\frac{2g+2}{18g+15}.$$

The ordinary generating function is

$$\begin{aligned}
F_g(x) = x + x^g + x^{8g+7} &+ \frac{(x^{4g+3} + x^{6g+5})(1-x^{g+1})}{1-x}\\
&+ \frac{x^M(x^{4g+2} + x^{6g+4})(1-x^{g+1})}{(1-x)(1-x^M)}, \qquad M = 18g+15.
\end{aligned}$$

8 Conclusion and Open Problems

We have given exact eventually periodic classifications for the regular rank-r bands, their boundary points, and a dense portion of each following transition window. Together these regimes classify, in every block of $2h$ third-seed values, a proportion tending to $3/4$ as h grows. The rank-one regular theorem resolves [1, Conjecture 6] and includes the adjacent endpoint $d = g$. The proofs combine the seeded safety-and-coverage criterion with restricted sums of integer intervals and modular tail-safety arguments.

The unclassified part of each transition window is

$$0 \leq t < \left\lceil \frac{h+2}{2} \right\rceil, \qquad t = h-1,\ h,$$

where $D = 2rh + t$. This leaves $\lceil (h+2)/2 \rceil + 2$ transition values in each recurrent block. Natural next questions are whether the sparse transition range has a uniform finite-packet description, whether the displayed eventual periods are minimal in every family, and whether a single master theorem can cover the regular, boundary, dense, and sparse regimes.

References

[1] W. Bosma, R. Bruin, R. Fokkink, J. Grube, A. Reuijl, and T. Tromp, *Using Walnut to solve problems from the OEIS*, arXiv:2503.04122, 2025, `https://arxiv.org/pdf/2503.04122`.